Amogh Gopalakrishna Nayak

Software Engineer | Boston, MA | nayak.amo@northeastern.edu | 857-339-8432 |
https://linkedin.com/in/amogh-nayak | https://github.com/amoghnayak07 | https://amoghnayak07.github.io/

Summary

Software engineer with 4+ years building distributed backend systems, high-throughput APIs, and AI-powered applications on AWS. Designed microservices handling 100K+ daily requests, shipped production LLM integrations, and managed Linux-based cloud infrastructure across multiple products.

Experience

Founder | BuRP – Business Resource Planning Nov 2023 - Aug 2024

- Built a multi-tenant ERP SaaS platform using microservices architecture with Node.js services and React components on AWS.
- Cut API response latency by 30% using async background processing and AWS ALB to distribute traffic across service instances.
- Automated Linux EC2 administration; managed Docker containers via SSH and configured cron-based S3 backup pipelines.
- Reduced deployment time by 60% by building Dockerized CI/CD pipelines with automated production rollouts on AWS.
- Generated $60K+ revenue in 3 months by using customer usage data to prioritize and ship high-impact features.

Software Development Engineer | LetsVenture Pvt. Ltd. June 2021 - October 2023

- Built REST APIs handling 100K+ daily requests across distributed microservices with optimized database query patterns.
- Built event-driven AWS Lambda pipelines to decouple async background processing from the main request lifecycle.
- Managed Linux EC2 deployments; administered Docker containers via SSH and coordinated zero-downtime production rollouts.
- Improved platform performance by 22% by configuring AWS ALB, CDN caching, and data access optimizations under peak traffic.
- Improved system reliability by 18% by implementing monitoring, logging, and alerting using AWS CloudWatch.
- Built async notification and event workflows using AWS SNS and SQS to decouple cross-system integrations for CRM, payments, and e-signature APIs.

Full-Stack Developer | Praan-e-Rakshak October 2020 - May 2021

- Reduced emergency response time by 50% by building a real-time MERN platform with WebSockets and live dashboards.
- Improved real-time coordination across rescue teams by integrating Google Maps APIs for live tracking and incident mapping.
- Enabled tracking of 500+ rescue cases by designing responsive dashboards for real-time case status and responder updates.

Full-Stack Development Intern | ResoluteAI July 2020 - October 2020

- Delivered a production-ready MVP for 100+ pilot users by developing React interfaces and backend integrations ahead of schedule.
- Improved platform security and access control by implementing authentication and role-based authorization using Firebase Auth.
- Reduced patient no-shows by 25% by integrating Zoom scheduling workflows and automated session management.
- Increased frontend reliability to 85% test coverage by implementing automated testing with Jest and React Testing Library.
- Improved usability of patient-facing workflows by redesigning interfaces for clearer navigation and accessibility.

Projects

Relay-Intent – Real-Time Chat App with Embedded AI Agent | (*Flask, React, Firebase, Google Gemini*)

- Built a context-aware AI agent by filtering and chunking chat history before each Gemini API call to control token usage.
- Implemented an observability layer tracking token consumption, latency, and agent decision paths across LLM calls.
- Designed a modular agent orchestration layer with extensible tool integration for natural-language command execution.
- Secured the app with cookie-based JWT authentication and CSRF protection across the React frontend and Flask backend.

Game Creator (AI Dungeon Master) | (*React, FastAPI, PostgreSQL*)

- Designed structured prompt templates with output schemas and a parsing layer to validate LLM responses into typed game objects.
- Built a FastAPI backend managing game generation, session state, and turn-by-turn metadata persisted in PostgreSQL.
- Developed a responsive React interface enabling real-time player interaction with AI-generated story environments.

Education

Master of Science in Computer Science (GPA: 3.63/4) | Northeastern University, Boston, USA Expected May 2026
Courses: Algorithms, Web Development, Foundations of AI, App Dev, Distributed Systems, Database Management System

Bachelor of Engineering in Telecommunication Engineering | DSCE, Bangalore, India August 2016 - August 2020
Courses: C Programming, Python, Java, C++

Technical Skills

Languages:	Python, JavaScript, TypeScript, Java, C/C++, HTML/CSS
Frameworks & Libraries:	Django, Flask, FastAPI, Express.js, Node.js, React, React Native, TensorFlow, PyTorch
Databases & DevOps:	MongoDB, MySQL, PostgreSQL, Firebase, AWS (EC2, Lambda, ALB, CloudWatch, S3, SNS, SQS), Docker, Bitbucket Pipelines, Nginx
ML:	RAG Pipelines, LLM Integration, Agent Orchestration, Prompt Engineering, Context Window Management